# Vinayak Kamat

Machine Learning Engineer | Production ML, Model Monitoring and Financial AI

Mumbai, India vinayakkamat001@gmail.com 072081 14510 vinayak-kamat-81b8901a0

vinayakkamatcodes

## Summary

Machine Learning Engineer with 1 year 8 months at KAN Innovations building, calibrating, and deploying production ML models in Python and PyTorch. Experience with real-time model-output monitoring, ONNX inference optimization, MLflow-based experiment tracking, and AWS deployment. Built Aethelgard, a real-time gateway that scores LLM outputs for risk and streams metrics to Datadog. Pursuing an MSc in Financial Engineering.

## Experience

**KAN Innovations**, Junior AI/ML Engineer — Mumbai, Maharashtra
May 2025 – present
1 year 6 months

- Improved model calibration so prediction confidence better reflects real accuracy, making production predictions more reliable
- Converted PyTorch models to ONNX to speed up inference in production
- Deployed ML services on AWS (Lambda, ECR, S3, DynamoDB) and integrated models into production pipelines
- Tracked experiments and versioned models with MLflow so results were reproducible and auditable
- Built and evaluated Retrieval-Augmented Generation (RAG) pipelines, measuring output quality across retrieval and generation

**KAN Innovations**, ML Engineer Intern — Mumbai, Maharashtra
Feb 2025 – May 2025
3 months

- Built a PyTorch ulcer detection model reaching about 85% accuracy
- Improved the performance of an eye posture detection model

## Projects

**Aethelgard**
Real-time AI governance gateway that scores LLM outputs for hallucination and risk.

- Built a FastAPI gateway that analyzes model outputs in real time, using a secondary model for semantic risk scoring
- Streamed risk metrics to Datadog for continuous monitoring of model output quality
- Designed automated detection logic to flag unreliable outputs before they reach users

**Financial AI Agent (Demo)**
Prototype agent for financial data retrieval and reasoning in Python.

- Built modular retrieval and reasoning pipelines for financial insights
- Implemented query routing and response refinement in Python

## Skills

**ML and Frameworks:** Python, PyTorch, Transformers, spaCy, OpenCV, ONNX, model calibration

**MLOps and Monitoring:** MLflow (experiment tracking, model versioning), Datadog (metrics streaming), Docker

**Backend and Cloud:** FastAPI, Flask, REST APIs, AWS (Lambda, ECR, S3, DynamoDB)

**Data:** Pandas, NumPy

## Education

**MSc** **WorldQuant University**, Financial Engineering — Remote, Washington DC
Sept 2025 – May 2027

**B.Tech Punjab Technical University**, Computer Science — Punjab, India
Sept 2021 – May 2025

## Achievements

Qualified the Graduate Aptitude Test in Engineering (GATE)